SANTHOSH R

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Full Stack Engineer ◇ React + TypeScript ◇ Node.js ◇ Linux ◇ Real-Time & Embedded Systems

SUMMARY

Full stack engineer, 2.5+ years shipping a production **React + TypeScript** front end and **Node.js/Express** backend for industrial power-monitoring hardware — real-time **Socket.IO** APIs, **JWT** auth with role-based access, and packaged **ARM Linux** deployment.

SKILLS

- **Languages:** TypeScript, JavaScript (ES6+), Python, C, C++, SQL
- **Frontend & Mobile:** React 18/19, Redux Toolkit, Zustand, React Router, Tailwind CSS, Recharts, Vite, React Native CLI
- **Backend:** Node.js, Express 5, REST API design, Socket.IO / WebSockets, JWT (access + refresh), RBAC, Swagger/OpenAPI, Winston
- **Python:** FastAPI, Flask, SQLAlchemy, Pydantic, Selenium, Tkinter, pandas / scikit-learn (gradient-boosted models), Docker & PyInstaller packaging
- **Databases & Architecture:** SQLite, Firebase/Firestore + security rules, PostgreSQL (Supabase, migrations, RLS), MySQL, MongoDB; system design, real-time data pipelines, API contract design
- **Linux & DevOps:** Ubuntu and embedded ARM Linux, Bash scripting, systemd units, nginx (reverse proxy + static hosting), udev automount, journald, Git, GitLab, Jest, `pkg` (ARM64 binaries), Vercel, Postman
- **Domain:** Industrial IoT, CAN / Modbus / I2C, embedded Linux touch panels, real-time device telemetry

WORK EXPERIENCE

Senior Software Engineer — Full Stack — 06/2025 – Present

Delta Electronics India, Bengaluru

- Sole developer of the **PDU Gen-3 platform**: a three-application system — an embedded React touch-panel UI, a browser-based admin configurator, and a Node.js/Express microservice — shipped on customer-facing Power Distribution Units.
- Designed and built the backend from scratch in **TypeScript**: 17 route modules covering measurement, alarms, calibration, event/data logs, configuration, user management and system control, documented with **Swagger/OpenAPI**.
- Implemented **JWT authentication** with short-lived access tokens and httpOnly refresh cookies, plus **role-based access control** where permissions are re-resolved on every refresh so an admin's revocation takes effect immediately on live sessions.
- Built a **real-time energy sampler** that runs independently of the UI — samples power every 5s into a rolling one-hour window and mirrors it to disk, so data collection never pauses on navigation and survives a backend restart.
- Replaced per-page config fetching with a **boot-time bootstrap and API self-test**, surfacing a live "backend disconnected" indicator by reusing the existing Socket.IO signal — no extra heartbeat endpoint needed.
- Delivered a **Vite + React + Tailwind** admin console for remote configuration, with permission-driven navigation.
- Automated device deployment with a **Bash self-extracting installer** that ships the packaged **ARM64 Linux binary** and both web builds, provisions **systemd** units and an **nginx** reverse proxy, and configures **udev**-driven USB automount — with versioned OTA updates and rollback backups.
- Added structured **journald** logging, USB import/export and DB backup/restore for field servicing.

Graduate Engineer Trainee — Frontend — 08/2024 – 05/2025

Delta Electronics India, Bengaluru

- Built the frontend of a customer-facing **PDU touch panel** in **React and TypeScript**, overhauling a legacy layout into a responsive interface: architected state with **Redux** and built a reusable component library from **Figma** designs, within hardware constraints (fixed resolution, touch-only input, custom on-screen keyboard).

Software Engineer Intern — 01/2024 – 07/2024

Delta Electronics India, Bengaluru

- Worked with **CAN, Modbus and I2C** protocols in industrial hardware; gained exposure to **Linux CLI** tooling and **GitLab**, and reworked PDU UI/UX mockups in **Figma**.

PROJECTS

Jourvia — Travel Companion App *React Native 0.85 (New Architecture), TypeScript, Firebase, Zustand Android & iOS*

- Cross-platform trip planner on the **React Native CLI** (no Expo): trip CRUD, Google Maps route distance, weather, friends via share IDs, and trip sharing with owner / editor permission tiers.
- Implemented **real-time live-location sharing** between trip companions on Firestore, plus an offline-first caching layer over **Zustand** stores so trips stay readable and editable without a connection.
- Wrote the **Firestore security rules** that moved authorization out of client JavaScript and onto the server, and covered utilities, offline sync and UI flows with a **Jest** test suite.

Habex — Expense, Habit & Investment Tracker *React, TypeScript, Supabase (Postgres), shadcn/ui, Tailwind* habex.santhu.site · GitHub

- Live personal-finance app on my own domain: expense entry with recurring rules and monthly views, habit tracking with a 21-day challenge, shared investment groups, and an admin dashboard.
- Backed by **Supabase Postgres** with 14+ versioned **SQL migrations** and three **edge functions** (PIN login, admin user management, activity audit) keeping privileged operations off the client.

FraudShield — Accounts-Payable Fraud Detection API *Python, FastAPI, SQLAlchemy, Pydantic*

- REST service that scores supplier invoices for fraud risk, with reviewer and admin workflows and auto-generated **OpenAPI** docs.
- Built a configurable **rules engine** — amount thresholds, per-vendor invoice velocity over a rolling window, and repeated round-amount detection — feeding a scoring service; paired with a **gradient-boosted** model trained on a labelled fraud dataset.

Time Tracker — Desktop Work-Hour Automation *Python, Tkinter, Selenium, PyInstaller*

- Desktop app that automates weekly timesheet entry: drives an attendance portal with **Selenium**, paging week by week and waiting on the rendered date before scraping each day's records rather than relying on fixed sleeps.
- Runs scraping off the UI thread and marshals results back to Tk's main loop, coalesces per-keystroke writes into one debounced save, and ships as a standalone Windows executable via **PyInstaller**.

EDUCATION

Siddaganga Institute of Technology — B.E. Electronics & Instrumentation, CGPA 8.94 **Dec 2020 – Jun 2024**

Vidyavahini PU College — Class 11 & 12, Pre-University (PCMB), 88.56% May 2018 – Jul 2020

Vivekananda Vidya Mandira School — Class X (SSLC), 78.64% May 2017 – May 2018

ADDITIONAL

Hack-o-Smart — 24-hour Tumkur Smart City hackathon; built a district-level education platform surfacing seat availability and fee details. ⋄ **Certifications:** SQL (CodeChef), Digital Skills (HackerRank) ⋄ **Leadership:** NCC Rank Holder, 4 Kar Tumkur Battalion